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April 2023

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Activity distribution among the hamstring muscles during high-speed running: A descriptive multichannel surface EMG study. Scandinavian Journal of Medicine and Science in Sports, Ahead of Print: 1-22, 2023.

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London International Consensus and Delphi study on hamstring injuries Part 1: Classification. British Journal of Sports Medicine, Ahead of Print: 1-14, 2023.

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London International Consensus and Delphi study on hamstring injuries Part 3: Rehabilitation, running and return to sport. British Journal of Sports Medicine, Ahead of Print: 1-17, 2023.

Reduced match exposure in the previous 2 matches accounts for hamstring muscle injury incidence in professional football players. Sports Health, Ahead of Print: 1-6, 2023.

Risk factors for hamstring muscle injury in male elite football: Medical expert experience and conclusions from 15 European Champions League clubs. BMJ Open Sport & Exercise Medicine, 9(e001461): 1-9, 2023.

The effectiveness of two comprehensive recovery protocols on performance and physiological measures in elite soccer players: A parallel group-randomized trial. International Journal of Sports Science & Coaching, Ahead of Print: 1-11, 2023.

The impact of nonoperative hip and core injuries on National Football League athlete performance. The Physician and Sportsmedicine, Ahead of Print: 1-7, 2023.

Training load and acute performance decrements following different training sessions. International Journal of Sports Physiology and Performance, Ahead of Print: 1-9, 2023.

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A qualitative study of 11 world-class team sport athletes' experiences answering subjective questionnaires: A key ingredient for 'visible' health and performance monitoring? Sports Medicine, Ahead of Print: 1-16, 2023.

Background: Athlete monitoring trends appear to be favoring objective over subjective measures. One reason of potentially several is that subjective monitoring affords athletes to give dishonest responses. Indeed, athletes have never been systematically researched to understand why they are honest or not.

Objective: Because we do not know what motivates professional athletes to be honest or not when responding to subjective monitoring, our objective is to explore the motives for why the athlete may or may not respond honestly.

Methods: A qualitative and phenomenological approach was used, interviewing 11 world-class team-sport athletes (five women, six men) about their experiences when asked to respond to subjective monitoring questionnaires. Interview transcripts were read in full and significant quotations/statements extracted. Meanings were formulated for each interviewees' story and assigned codes. Codes were reflected upon and labelled as categories, with similar categories grouped into an overall theme. Themes were examined, articulated, re-interpreted, re-formulated, and written as a thematic story, drawing on elements reported from different athletes creating a blended story, allowing readers a feel for what it is like to live the experience.

Results: Overall, four key themes emerged: (i) pursuit of the ideal-self, (ii) individual barriers to athlete engagement, (iii) social facilitators to athlete engagement; and (iv) feeling compassion from performance staff.

Conclusions: Our main insight is that athletes' emotions play a major role in whether they respond honestly or not, with these emotions being driven at least in part by the performance staff asking the questions.

Key Points

- Team-sport athletes are initially attracted to engaging in subjective monitoring with curiosity and openness that it might help them to become a better athlete and are willing to consider answering honestly; however, through their behaviors and actions, performance staff can invoke either positive or negative emotions in the players, which ultimately drives their level of honesty.
- Despite common sense dictating that performance staff should cultivate a trusting and transparent relationship with athletes as well as facilitate an engaging environment, our study

shows the power that performance staff actually have to either positively or negatively affect their athletes and the consequences for subjective monitoring.

- To claim that subjective monitoring is a 'waste of time' because athletes are not honest without demonstrating this claim through empirically derived evidence has hindered the scientific investigation of "why" athletes are honest or not. Performance staff should reflect on their own behaviors and actions when delivering subjective monitoring.
- The findings of this study bring into question the concept of 'invisible' monitoring as it is currently defined as this contradicts what athletes actually want from athlete health and performance monitoring, opening up new possibilities for 'visible' monitoring.

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Activity distribution among the hamstring muscles during high-speed running: A descriptive multichannel surface EMG study. Scandinavian Journal of Medicine and Science in Sports, Ahead of Print: 1-22, 2023.

Purpose: This study assessed activity distribution among the hamstring muscles during high-speed running. The objective was to compare within and between muscle activity, relative contribution and hip and knee joint angles at peak muscle activity during high-speed running.

Methods: Through multichannel electromyography, we measured muscle activity in male basketball players during high-speed running on a treadmill at 15 locations: five for biceps femoris long head, four for semitendinosus, six for semimembranosus. Muscle activity was calculated for each location within each hamstring muscle individually for each percent of a stride cycle.

Results: Twenty-nine non-injured basketball players were included (mean age: 17 ± 1 years; mass, 85 ± 9 kg; height, 193 ± 9 cm). Heterogeneous activity was found for all individual hamstring muscles across multiple events of the stride cycle. In the late-swing phase, muscle activity and relative contribution of the semimembranosus was significantly higher than of the semitendinosus. There was no significant difference in hip and knee joint angles at instant of peak muscle activity, assessed locally within individual hamstring muscles, as well as in general over the whole hamstring muscle.

Conclusion: Hamstring muscles were most active in the late-swing phase during high-speed running. In this phase, the semimembranosus was most active and the semitendinosus was least active. Within the biceps femoris long head, the most proximal region was significantly more active in the late-swing. phase, compared to other muscle regions. For each muscle and location, peak muscle activity occurred at similar hip and knee joint angles.

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Basics must improve to reduce the burden of hamstring muscle injuries. British Journal of Sports Medicine, Ahead of Print: 1-2, 2023.

Overview: Despite a variety of injury prevention programs and a rapid increase in research over the past decade, the incidence of hamstring muscle injuries (HSI) in many competitive sports has not decreased.1 Risk factors are multifactorial and may include anthropometry, game volume and intensity, competitive timetabling (congestion), length of career, international travel and duties and of course a history of prior injury and compliance to adequate rehabilitation. While some risk factors are not modifiable, inappropriate decision-making and/or non-adherence to rehabilitation practices may partially explain the increased incidence (and reoccurrence of HSI).2 Clinicians managing athletes with HSI require the best available evidence to support their management in optimizing return to play. Unfortunately, evidence in key areas remains weak. For example, criteria used to guide return to full sprinting do not encompass the differing roles of each hamstring muscle in sprinting. Furthermore, agreement is lacking on when to consider surgical intervention to restore connective tissue architecture, tension and reduce reinjury risk in severe HSI. Given the increasing incidence and burden of HSI, together with what appears to be suboptimal injury prevention and rehabilitation evidence, the sports and exercise medicine community should be asking 'have we even got the basics nailed down?'

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Do sports compression garments alter measures of peripheral blood flow? A systematic review with meta-analysis. Sports Medicine, Ahead of Print: 1-21, 2023.

Background: One of the proposed mechanisms underlying the benefits of sports compression garments may be alterations in peripheral blood flow.

Objective: We aimed to determine if sports compression garments alter measures of peripheral blood flow at rest, as well as during, immediately after and in recovery from a physiological challenge (i.e. exercise or an orthostatic challenge).

Methods: We conducted a systematic literature search of databases including Scopus, SPORTDiscus and PubMed/MEDLINE. The criteria for inclusion of studies were: (1) original papers in English and a peer-reviewed journal; (2) assessed effect of compression garments on a measure of peripheral blood flow at rest and/or before, during or after a physiological challenge; (3) participants were healthy and without cardiovascular or metabolic disorders; and (4) a study population including athletes and physically active or healthy participants. The PEDro scale was used to assess the methodological quality of the included studies. A random-effects meta-analysis model was used. Changes in blood flow were quantified by standardized mean difference (SMD) [±95% confidence interval (CI)].

Results: Of the 899 articles identified, 22 studies were included for the meta-analysis. The results indicated sports compression garments improve overall peripheral blood flow (SMD=0.32, 95% CI 0.13, 0.51, p= 0.001), venous blood flow (SMD=0.37, 95% CI 0.14, 0.60, p=0.002) and arterial blood flow (SMD=0.30, 95% CI 0.01, 0.59, p=0.04). At rest, sports compression garments did not improve peripheral blood flow (SMD=0.18, 95% CI − 0.02, 0.39, p=0.08). However, subgroup analyses revealed sports compression garments enhance venous (SMD=0.31 95% CI 0.02, 0.60, p=0.03), but not arterial (SMD=0.12, 95% CI− 0.16, 0.40, p=0.16), blood flow. During a physiological challenge, peripheral blood flow was improved (SMD=0.44, 95% CI 0.19, 0.69, p=0.0007), with subgroup analyses revealing sports compression garments enhance venous (SMD=0.48, 95% CI 0.11, 0.85, p=0.01) and arterial blood flow (SMD=0.44, 95% CI 0.03, 0.86, p=0.04). At immediately after a physiological challenge, there were no changes in peripheral blood flow (SMD=−0.04, 95% CI−0.43, 0.34, p=0.82) or subgroup analyses of venous (SMD =−0.41, 95% CI−1.32, 0.47, p=0.35) and arterial (SMD =0.12, 95% CI−0.26, 0.51, p=0.53) blood flow. In recovery, sports compression garments did not improve peripheral blood flow (SMD=0.25, 95% CI − 0.45, 0.95, p=0.49). The subgroup analyses showed enhanced venous (SMD=0.67, 95% CI 0.17, 1.17, p=0.009), but not arterial blood flow (SMD=0.02, 95% CI − 1.06, 1.09, p=0.98).

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Conclusions: Use of sports compression garments enhances venous blood flow at rest, during and in recovery from, but not immediately after, a physiological challenge. Compression-induced changes in arterial blood flow were only evident during a physiological challenge.

Key Points

- The results of this systematic review and meta-analysis suggest that sports compression garments enhance peripheral blood flow, particularly venous blood flow.
- Compression-induced improvements in venous blood flow were evident at rest, during and in recovery from a physiological challenge, but not immediately after a physiological challenge timepoint.
- Studies involving sports compression garments present moderate methodological quality because of the lack of blinding of participants and researchers.

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Effects of different hamstring eccentric exercise programs on preventing lower extremity injuries: A systematic review and meta-analysis. International Journal of Environmental Research and Public Health, 20(2057): 1-15, 2023.

Abstract - This systematic review and meta-analysis aims to investigate the effects and differences of various hamstring eccentric training protocols for the prevention of lower limb injuries, and we further propose a more refined hamstring eccentric training protocol for the prevention of lower limb injuries. A literature search for the effects of hamstring eccentric training on lower extremity sports injuries was conducted using the PubMed, Web of Science, and EMBASE databases, and the literature was searched covering the period from the date of the database's creation to 20 August 2022. A meta-analysis of the included literature was performed using R.4.21 for lower extremity injuries, injuries in various parts of the lower extremity, and subgroup analysis for exercise frequency, exercise cycle, and exercise population. A total of 23 randomized controlled trial (RCT) studies were found to be included in the meta-analysis, and 15 of these trials, totaling 14,721 patients, were determined to be included in the overall lower extremity injury prevention effect. The analysis showed that the implementation of a hamstring eccentric training program reduced lower extremity injuries by 28%, and it resulted in a 46% decrease in hamstring injury rate and a 34% decrease in knee injury rate. The subgroup analysis revealed that the frequency of exercise was most significant in the twice-a-week exercise group, that the exercise program was most effective in preventing injuries in the 21–30-week exercise period, and that the program was most effective in preventing injuries in elite athletes and amateur adult athletic populations, compared with adolescents.

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Exploring the acute cardiovascular effects of Floatation-REST. Frontiers in Neuroscience, Dec. 9: 1-16, 2022.

Abstract - The central nervous system (CNS) exerts a strong regulatory influence over the cardiovascular system in response to environmental demands. Floatation-REST (Reduced Environmental Stimulation Therapy) is an intervention that minimizes stimulation from the environment, yet little is known about the autonomic consequences of reducing external sensory input to the CNS. We recently found that Floatation-REST induces a strong anxiolytic effect in anxious patients while paradoxically enhancing their interoceptive awareness for cardiorespiratory sensations. To further investigate the physiologic nature of this anxiolytic effect, the present study measured acute cardiovascular changes during Floatation-REST using wireless and waterproof equipment that allowed for concurrent measurement of heart rate, heart rate variability (HRV), breathing rate, and blood pressure. Using a within-subjects crossover design, 37 clinically anxious participants with high levels of anxiety sensitivity and 20 non-anxious comparison participants were randomly assigned to undergo a 90-min session of either Floatation-REST or an exteroceptive comparison condition that entailed watching a relaxing nature film. Measures of state anxiety and serenity were collected before and after each session, while indices of autonomic activity were measured throughout each session. HRV was calculated using both time-series and frequency domain analyses. Linear mixed-effects modeling revealed a significant main effect of condition such that relative to the film condition, Floatation-REST elicited significant decreases ($p < 0.001$) in diastolic blood pressure, systolic blood pressure, breathing rate, and certain metrics of HRV including the standard deviation of the interbeat interval (SDNN), low-frequency HRV, and very low-frequency HRV. Heart rate showed a non-significant trend ($p = 0.073$) toward being lower in the float condition, especially toward the beginning of the session. The only metric that showed a significant increase during Floatation-REST was normalized high-frequency HRV ($p < 0.001$). The observed physiological changes were consistent across both anxious and non-anxious participants, and there were no significant group by condition interactions. Blood pressure was the only cardiac metric significantly associated with float-related reductions in state anxiety and increases in serenity. These findings suggest that Floatation-REST lowers sympathetic arousal and alters the balance of the autonomic nervous system toward a more parasympathetic state.

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Food for thought: Physiological considerations for nutritional ergogenic efficacy. Scandinavian Journal of Medicine and Science in Sports, Jan. 3: 1-14, 2023.

Abstract - Top-class athletes have optimized their athletic performance largely through adequate training, nutrition, recovery, and sleep. A key component of sports nutrition is the utilization of nutritional ergogenic aids, which may provide a small but significant increase in athletic performance. Over the last decade, there has been an exponential increase in the consumption of nutritional ergogenic aids, where over 80% of young athletes report using at least one nutritional ergogenic aid for training and/or competition. Accordingly, due to their extensive use, there is a growing need for strong scientific investigations validating or invalidating the efficacy of novel nutritional ergogenic aids. Notably, an overview of the physiological considerations that play key roles in determining ergogenic efficacy is currently lacking. Therefore, in this brief review, we discuss important physiological considerations that contribute to ergogenic efficacy for nutritional ergogenic aids that are orally ingested including (1) the impact of first pass metabolism, (2) rises in systemic concentrations, and (3) interactions with the target tissue. In addition, we explore mouth rinsing as an alternate route of ergogenic efficacy that bypasses the physiological hurdles of first pass metabolism via direct stimulation of the central nervous system. Moreover, we provide real-world examples and discuss several practical factors that can alter the efficacy of nutritional ergogenic aids including human variability, dosing protocols, training status, sex differences, and the placebo effect. Taking these physiological considerations into account will strengthen the quality and impact of the literature regarding the efficacy of potential ergogenic aids for top-class athletes.

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Individual variability is more important than analytical methods when calculating relative speed of beverage bioavailability. International Journal of Sport Nutrition and Exercise Metabolism, Ahead of Print: 1-10, 2023.

Abstract - Deuterium oxide (D2O) appearance in blood is a marker of fluid bioavailability. However, whether biomarker robustness (e.g., relative fluid delivery speed) is consistent across analytical methods (e.g., cavity ring-down spectroscopy) remains unclear. Fourteen men ingested fluid (6 ml/kg body mass) containing 0.15 g/kg D2O followed by 45 min blood sampling. Plasma (D2O) was detected (n = 8) by the following: isotope-ratio mass spectrometry after vapor equilibration (IRMS-equilibrated water) or distillation (IRMS-plasma) and cavity ring-down spectroscopy. Two models calculated D2O halftime to peak (t1/2max): sigmoid curve fit versus asymmetric triangle (TRI). Background (D2O) differed ($p < .001$, $\eta 2 = .98$) among IRMS-equilibrated water, IRMS-plasma, and cavity ring-down spectroscopy (152.2 ± 0.8, 147.2 ± 1.5, and 137.7 ± 2.2 ppm), but did not influence ($p > .05$) D2O appearance (Δppm), time to peak, or t1/2max. Stratifying participants based on mean t1/2max (12 min) into “slow” versus “fast” subgroups resulted in a 5.8 min difference ($p < .001$, $\eta 2 = .73$). Significant t1/2max model ($p = .01$, $\eta 2 = .44$) and Model × Speed Subgroup interaction ($p = .005$, $\eta 2 = .50$) effects were observed. Bias between TRI and sigmoid curve fit increased with t1/2max speed: no difference ($p = .75$) for fast (9.0 min vs. 9.2 min, respectively) but greater t1/2max ($p = .001$) with TRI for the slow subgroup (16.1 min vs. 13.7 min). Fluid bioavailability markers are less influenced by which laboratory method is used to measure D2O as compared with the individual variability effects that influence models for calculating t1/2max. Thus, TRI model may not be appropriate for individuals with slow fluid delivery speeds.

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London International Consensus and Delphi study on hamstring injuries Part 1: Classification. British Journal of Sports Medicine, Ahead of Print: 1-14, 2023.

Abstract - Muscle injury classification systems for hamstring injuries have evolved to use anatomy and imaging information to aid management and prognosis. However, classification systems lack reliability and validity data and are not specific to individual hamstring muscles, potentially missing parameters vital for sport-specific and activity specific decision making. A narrative evidence review was conducted followed by a modified Delphi study to build an international consensus on best-practice decision-making for the classification of hamstring injuries. This comprised a digital information gathering survey to a cohort of 46 international hamstring experts (sports medicine physicians, physiotherapists, surgeons, trainers and sports scientists) who were also invited to a face-to-face consensus group meeting in London. Fifteen of these expert clinicians attended to synthesize and refine statements around the management of hamstring injury. A second digital survey was sent to a wider group of 112 international experts. Acceptance was set at 70% agreement. Rounds 1 and 2 survey response rates were 35/46 (76%) and 99/112 (88.4%) of experts responding. Most commonly, experts used the British Athletics Muscle Injury Classification (BAMIC) (58%), Munich (12%) and Barcelona (6%) classification systems for hamstring injury. Issues identified to advance imaging classifications systems include: detailing individual hamstring muscles, establishing optimal use of imaging in diagnosis and classification, and testing the validity and reliability of classification systems. The most used hamstring injury classification system is the BAMIC. This consensus panel recommends hamstring injury classification systems evolve to integrate imaging and clinical parameters around: individual muscles, injury mechanism, sporting demand, functional criteria and patient-reported outcome measures. More research is needed on surgical referral and effectiveness criteria, and validity and reliability of classification systems to guide management.

Key points

⇒ While classification systems exist for hamstring injuries (HSIs) and encompass anatomical and imaging criteria, current classification systems are not specific to individual (hamstring) muscles.

⇒ Classification systems have evolved to include the specific anatomical tissue (ie, muscle, myotendinous, tendon) as well as severity of injury gradings, and some include the mechanism of injury and athlete factors.

⇒ Clinicians most commonly use the British Athletics Muscle Injury Classification (BAMIC) system, with Munich and Barcelona systems also used for the classification of HSI.

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⇒ This expert panel recommends MRI as the imaging of choice for diagnosis with few panelists prioritizing diagnostic ultrasound. Neither modality is recommended as a means of monitoring rehab progression or deciding on readiness to return to sport.

⇒ Experts agree classification systems for HSI should evolve to include parameters around: individual hamstring muscles, intramuscular injuries, mechanism of injury, sporting demand, functional criteria and patient-reported outcome measures.

⇒ There is a need for more research into criteria that determine the need for surgical intervention.

⇒ There is a need for more research into the effectiveness of classification systems to prognosticate and guide treatment decision making.

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London International Consensus and Delphi study on hamstring injuries Part 2: Operative management. British Journal of Sports Medicine, Ahead of Print: 1-13, 2023.

Abstract - The key indications for surgical repair of hamstring injuries (HSIs) remain unclear in the literature due to a lack of high-level evidence and expert knowledge. The 2020 London International Hamstring Consensus meeting aimed to highlight clear surgical indications and to create a foundation for future research. A literature review was conducted followed by a modified Delphi process, with an international expert panel. Purposive sampling was used with two rounds of online questionnaires and an intermediate round involving a consensus meeting. The initial information gathering (round 1) questionnaire was sent to 46 international experts, which comprised open ended questions covering decision-making domains in HSI. Thematic analysis of responses outlined key domains, which were evaluated by a smaller international subgroup (n=15) comprising clinical academic sports medicine physicians, physiotherapists and orthopaedic surgeons in a consensus meeting. After group discussion of each domain, a series of consensus statements were prepared, debated and refined. A round 2 questionnaire was sent to 112 international hamstring experts to vote on these statements and determine level of agreement. The consensus threshold was set a priori at 70% agreement. Rounds 1 and 2 survey respondents were 35/46 (76%) and 99/112 (88.4%), respectively. The consensus group agreed that the indications for operative intervention included: gapping at the zone of tendinous injury (87.2% agreement) and loss of tension (70.7%); symptomatic displaced bony avulsions (72.8%); and proximal free tendon injuries with functional compromise refractory to non-operative treatment (72.2%). Other important considerations for operative intervention included: the demands of the athlete/patient and the expected functional outcome (87.1%) based on the anatomy of the injury; the risk of functional loss/ performance deficit with non-operative management (72.2%); and the capacity to restore anatomy and function (87.1%). Further research is needed to determine whether surgery can reduce the risk of reinjury as consensus was not reached within the whole group (48.2%) but was agreed by surgeons (70%) in the cohort. The consensus group did not support the use of corticosteroids or endoscopic surgery without further evidence. These guidelines will help standardize treatment of HSIs, specifically the indications and decision-making for surgical intervention.

Key messages

⇒ Indications for operative intervention of hamstring injuries (HSIs) include: gapping and loss of tension at the zone of tendinous injury; symptomatic displaced bony avulsions; and proximal free tendon injuries with functional compromise not responsive to non-operative treatment.

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⇒ Surgical intervention has the capacity to restore anatomy and function and may reduce the risk of injury recurrence.

⇒ Elite level athletes with severe-grade musculotendinous junction and distal HSIs demanding return to previous sporting level and a low risk of reinjury, merit surgical consideration as well as the non-surgical rehabilitation to achieve their specific demands.

⇒ Further research is warranted to refine criteria for surgical intervention and the decision-making process.

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London International Consensus and Delphi study on hamstring injuries Part 3: Rehabilitation, running and return to sport. British Journal of Sports Medicine, Ahead of Print: 1-17, 2023.

Abstract - Hamstring injuries (HSIs) are the most common athletic injury in running and pivoting sports, but despite large amounts of research, injury rates have not declined in the last 2 decades. HSI often recur and many areas are lacking evidence and guidance for optimal rehabilitation. This study aimed to develop an international expert consensus for the management of HSI. A modified Delphi methodology and consensus process was used with an international expert panel, involving two rounds of online questionnaires and an intermediate round involving a consensus meeting. The initial information gathering round questionnaire was sent to 46 international experts, which comprised open-ended questions covering decision-making domains in HSI. Thematic analysis of responses outlined key domains, which were evaluated by a smaller international subgroup (n=15), comprising clinical academic sports medicine physicians, physiotherapists and orthopaedic surgeons in a consensus meeting. After group discussion around each domain, a series of consensus statements were prepared, debated and refined. A round 2 questionnaire was sent to 112 international hamstring experts to vote on these statements and determine level of agreement. Consensus threshold was set a priori at 70%. Expert response rates were 35/46 (76%) (first round), 15/35 (attendees/invitees to meeting day) and 99/112 (88.2%) for final survey round. Statements on rehabilitation reaching consensus centered around: exercise selection and dosage (78.8%–96.3%agreement), impact of the kinetic chain (95%), criteria to progress exercise (73%–92.7%), running and sprinting (83%–100%) in rehabilitation and criteria for return to sport (RTS) (78.3%–98.3%). Benchmarks for flexibility (40%) and strength (66.1%) and adjuncts to rehabilitation (68.9%) did not reach agreement. This consensus panel recommends individualized rehabilitation based on the athlete, sporting demands, involved muscle(s) and injury type and severity (89.8%). Early-stage rehab should avoid high strain loads and rates. Loading is important but with less consensus on optimum progression and dosage. This panel recommends rehabilitation progress based on capacity and symptoms, with pain thresholds dependent on activity, except pain-free criteria supported for sprinting (85.5%). Experts focus on the demands and capacity required for match play when deciding the rehabilitation end goal and timing of RTS (89.8%). The expert panelists in this study followed evidence on aspects of rehabilitation after HSI, suggesting rehabilitation prescription should be individualized, but clarified areas where evidence was lacking. Additional research is required to determine the optimal load dose, timing and criteria for HSI rehabilitation and the monitoring and testing metrics to determine safe rapid progression in rehabilitation and safe RTS. Further research would benefit optimizing: prescription of running and sprinting, the application of adjuncts in rehabilitation and treatment of kinetic chain HSI factors.

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Reduced match exposure in the previous 2 matches accounts for hamstring muscle injury incidence in professional football players. Sports Health, Ahead of Print: 1-6, 2023.

Background: Hamstring strain injuries are one of the most prevalent injuries in football (soccer). We examined the influence of accumulated match-play exposure on the occurrence of hamstring strain injury in professional football from 2 teams (Spanish 1st Division, LaLiga) over 3 seasons, and determined specific cut-off points as indicators of injury risk.

Hypothesis: Overloaded players would be more likely to sustain a hamstring injury.

Study Design: Prospective, controlled, observational study.

Level of Evidence: Level 2b.

Methods: Playing time, total running distance, and high-speed running (>24 km/h) distance during official matches of players that sustained a hamstring injury were compared with uninjured, paired controls. Cumulative playing time and running performance of 4 matches before the injury was computed. Relative risk (RR) of injury occurrence was estimated by generalized estimating equations. Diagnostic accuracy was determined by receiver operating characteristics and the area under the curve.

Results: Thirty-seven hamstring strain injuries occurred, representing 23 ± 18 absence days per injury. Thirty-seven controls (uninjured players) were used as comparators. Low match-play exposures during 1 and 2 matches before injury were likely to explain injury occurrence (RR: 14-53%; $P < 0.01$). Metrics from the match before the hamstring muscle strain demonstrated the best accuracy to predict injury occurrence: high-speed running distance ≤328 m (sensitivity, 64%; specificity, 84%), playing time ≤64 min (sensitivity, 36%; specificity, 97%), and running distance ≤5.8 km (sensitivity, 39%; specificity, 97%).

Conclusion: Relatively reduced competitive exposure in the previous 2 matches was associated with higher hamstring injury risk in professional football players.

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Risk factors for hamstring muscle injury in male elite football: Medical expert experience and conclusions from 15 European Champions League clubs. BMJ Open Sport & Exercise Medicine, 9(e001461): 1-9, 2023.

Objectives: To describe the perceived importance of suggested hamstring injury risk factors according to chief medical officers (CMOs) of European male professional football clubs. A secondary objective was to compare if these perceptions differed between teams with a lower than-average hamstring injury burden and teams with a higher than average hamstring injury burden.

Methods: First, CMOs of 15 European professional male football clubs were asked to suggest risk factors for hamstring injury in their club. The perceived importance of the suggested risk factors was then rated by all participants on a 5-graded Likert scale. Participating teams were divided in two groups depending on their hamstring injury burden during the 2019/2020 and 2020/2021 seasons. The LOW group consisted of seven teams that had a lower-than-average hamstring injury burden. The HIGH group consisted of eight teams that had a higher-than-average hamstring injury burden.

Results: Twenty-one risk factors were suggested. The majority were extrinsic in nature, associated with coaching staff, team or club rather than players themselves. 'Lack of communication between medical staff and coaching staff' had the highest average importance (weighted average=3.7) followed by 'Lack of regular exposure to high-speed football during training sessions' (weighted average=3.6). The HIGH group perceived the player factors fatigue and wellness as more important than the LOW group.

Conclusion: According to CMOs recruited in this study, most risk factors for hamstring injuries are extrinsic and associated with the club and coaching staff, and not the players themselves.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Hamstring injury is the most common injury diagnosis in professional football.

⇒ While programs to prevent hamstring injuries have been shown to be effective, no reduction in the rate of hamstring injuries has been observed in male professional football.

WHAT THIS STUDY ADDS

⇒ Several risk factors may contribute to hamstring injuries ⇒ According to the chief medical officer's, the majority of these are extrinsic (ie, associated with the club and coaching staff) rather than intrinsic (associated with the players themselves).

⇒ Most of the risk factors proposed are associated with factors controlled by coaches such as excessive training, too many matches and overloading with subsequent increases in fatigue or poor training leading to undertraining and muscular dysfunction.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ This study highlights the responsibility of the club and coaching staff in reducing the risk for hamstring injury in professional football. A better understanding of the importance of communication between medical and coaching staff, as well as improvement in load management and training content during the football season could potentially lead to a fall in the rate of hamstring injuries among professional players.

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The effectiveness of two comprehensive recovery protocols on performance and physiological measures in elite soccer players: A parallel group-randomized trial. International Journal of Sports Science & Coaching, Ahead of Print: 1-11, 2023.

Abstract - Elite soccer players consistently report using several recovery methods after practices and games. However, it is unclear how their subsequent performance could be enhanced using broad protocols. The aim of this study was to compare the effectiveness of two different comprehensive recovery protocols in physiological, neuromuscular, and perceptual outcomes. Eighteen Mexican National Team players (19.56±0.62 years) were recruited. Using a randomized parallel group design, players followed one of two recovery protocols: (a) carbohydrate and protein shake, foam roller, cold-water immersion, and tart cherry juice concentrate (n=9) or (b) carbohydrate and protein shake, stretching, and intermittent cold-water immersion (n=9) following the completion of an unofficial game and the day after. Muscular creatine kinase, countermovement jump, hamstring maximal voluntary contraction, perceived recovery, and muscle soreness were assessed before, immediately after, and 20-44-68 h post-game. Significant effects (set at $p \leq 0.05$) of time were present in both interventions. Muscular creatine kinase was normalized entirely at 68 h post-game, while neuromuscular and perceptual outcomes were homogenized at 44 h. No statistically significant effects between protocols were found. Still, the interaction effects showed changes only in the group using protocol (a) at post-game and 20 h after in neuromuscular and muscle soreness. There were large correlations between muscular creatine kinase and accelerations, decelerations, sprints, and high-speed running distances. In conclusion, the interventions are equally effective for improving physiological, neuromuscular, and perceptual outcomes. Thus, elite soccer players may benefit from different combinations of methods after practices or games to obtain positive effects on recovery after them.

Sports Science Research Update

The impact of nonoperative hip and core injuries on National Football League athlete performance. The Physician and Sportsmedicine, Ahead of Print: 1-7, 2023.

Objectives: Hip and core injuries are common in National Football League (NFL) athletes; however, the impact following injury remains unclear. The goal of this manuscript was to determine the impact of nonoperative hip and core injuries on return to play and performance.

Methods: NFL athletes who sustained a hip or core injury treated nonoperatively between 2010 and 2016 were identified. Offensive and defensive power ratings were calculated for each player's injury season and two seasons before and after to assess longitudinal impact. A matched control group without an identified hip and/or core injury was assembled for comparison.

Results: A total of 41 offensive and 71 defensive players with nonoperative hip or core injury were analyzed. All athletes returned to play; offensive and defensive players missed 4.0 ± 5.2 and 3.1 ± 2.6 games after injury, respectively. Offensive players played fewer cumulative career games returning from core injury versus hip (23.5 ± 20.6 vs 41.0 ± 26.4). Defensive players played fewer games (58.1 ± 41.1 versus 37.4 ± 27.1, $p < 0.05$) with lower defensive power rating (133.9 ± 128.5 versus 219.8 ± 212.2, $p < 0.05$) cumulatively after hip or core injury. Additionally, 2 years following injury, defensive players played fewer games compared to controls (9.5 ± 7.0 versus 10.9 ± 6.8, $p < 0.05$). Following hip injury specifically, NFL defenders played fewer games (39.8 ± 27.9 vs 61.9 ± 38.8; $p < 0.05$) and had a lower defensive power rating (145.9 ± 131.7 vs 239.0 ± 205.9; $p < 0.05$) compared to before injury.

Conclusion: Overall, NFL players return to play following nonoperative hip and core injuries. Defensive players played in fewer games following hip or core injury compared to controls; offensive players were unaffected. Hip injuries have a greater impact on performance compared to core injuries in defensive athletes; offensive players played fewer games upon return from core injury.

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Training load and acute performance decrements following different training sessions. International Journal of Sports Physiology and Performance, Ahead of Print: 1-9, 2023.

Purpose: To examine the differences in training load (TL) metrics when quantifying training sessions differing in intensity and duration. The relationship between the TL metrics and the acute performance decrement measured immediately after the sessions was also assessed.

Methods: Eleven male recreational cyclists performed 4 training sessions in a random order, immediately followed by a 3-km time trial (TT). Before this period, participants performed the time TT in order to obtain a baseline performance. The difference in the average power output for the TTs following the training sessions was then expressed relative to the best baseline performance. The training sessions were quantified using 7 different TL metrics, 4 using heart rate as input, 2 using power output, and 1 using the rating of perceived exertion.

Results: The load of the sessions was estimated differently depending on the TL metrics used. Also, within the metrics using the same input (heart rate and power), differences were found. TL using the rating of perceived exertion was the only metric showing a response that was consistent with the acute performance decrements found for the different training sessions. The Training Stress Score and the individualized training impulse demonstrated similar patterns but overexpressed the intensity of the training sessions. The total work done resulted in an overrepresentation of the duration of training.

Conclusion: TL metrics provide dissimilar results as to which training sessions have higher loads. The load based on TL using the rating of perceived exertion was the only one in line with the acute performance decrements found in this study.